

Lesson 8 Proof by Contradiction

In Lesson 7, we learned proof by contrapositive. In this lesson, we study another indirect technique: **proof by contradiction**, which we introduce with an example.

Proposition 1 There is no largest integer.

Proof Suppose there is a largest integer, let's call it x .

Consider now another integer, $y = x + 1$. Then y is larger than x .

This contradicts the premise that x is a largest integer.

Therefore, there is no largest integer.

In a proof by contradiction, we assume that the assertion is false, or its negation is true. We then proceed until we reach a contradiction, that is, something false or absurd. This must mean that the negation is false, and the assertion is true. This argument is *reductio ad absurdum* in Latin: reduction to absurdity.

If we start a proof assuming that the assertion is false, then it is clear that we are doing a proof by contradiction, we may omit the last sentence in the above proof (blue). Some people omit more, ending the above proof with, "... y is larger than x , contradiction!" (or $\rightarrow\leftarrow$, which denotes contradiction). For a beginner, it is a good idea to state explicitly what the contradiction is.

Proof by contradiction is powerful. If you have a disagreement with someone and cannot convince them, try this: suppose they are right and then show that what absurdity follows, and at once they will be convinced that they are wrong!

It is possible that a direct proof and a proof by contradiction are equally good.

Proposition 2 Consecutive integers cannot be both even.

Proof 1 (direct) Let x be an arbitrary integer. If x is not even, then we are done. If x is even, then $x = 2n$ for $n \in \mathbb{Z}$, and $x + 1 = 2n + 1$ is odd, not even.

Proof 2 Suppose there are two consecutive even integers x and $x + 1$. Then $x = 2n$ and $x + 1 = 2m$ for some integers n and m , and $2n + 1 = 2m$, which implies that $1 = 2(m - n)$, and 1 is even, a contradiction!

Sometimes a proof by contradiction is clearly the way to go!

Proposition 3 $\sqrt{2}$ is irrational.

Proof Suppose that the contrary (i.e. the negation) is true, that is, $\sqrt{2}$ is rational. Then $\sqrt{2}$ can be written as an irreducible fraction of integers, i.e.

$$\sqrt{2} = \frac{m}{n}$$

where m and n do not have a common factor/divisor (other than 1 and -1).

It follows that $m^2/n^2 = 2$, which implies that $m^2 = 2n^2$.

Since n^2 is an integer, m^2 is even. By Proposition 2 in Lesson 7, m is even.

Since m is even, $m = 2k$ for some $k \in \mathbb{Z}$. Then $m^2 = (2k)^2 = 4k^2$.

Earlier, we have $m^2 = 2n^2$, so $2n^2 = 4k^2$ and $n^2 = 2k^2$.

Since k^2 is an integer, n^2 is even and n is even.

So both m and n are even, contradicting the fact that m/n is irreducible!

In the above proof, we have used the basic fact that any rational number can be written as an irreducible fraction of integers. The proof is a “classic” proof: people expect all math majors to know it! Following is another classic proof.

Proposition 4 There are infinitely many primes.

Proof We assume the **Fundamental Theorem of Arithmetic** (FTA):

Any integer $n \geq 2$ has a prime factor.

Suppose there are finitely many primes, call them $p_1 = 2, p_2, \dots, p_t$. Let $n = p_1 p_2 p_3 \dots p_t + 1$. Then $n \geq 2$ is an integer, but none of the primes divides n (there is always a remainder of 1), contradicting the FTA.

(We have used FTA without proof but we will prove FTA in Lesson 12.)

Many conditional statements $p \rightarrow q$ can be proved both ways: by contrapositive or by contradiction. The proofs are **NOT** exactly the same.

- Proof by contrapositive shows that $\sim q \rightarrow \sim p$ is true
- Proof by contradiction shows that the negation of $\sim(p \rightarrow q)$, that is, $p \wedge \sim q$, is false.

Proposition 5 Let a and b be real numbers. If $ab = 1$, then $a \leq 1$ or $b \leq 1$.

This is Proposition 3 in Lesson 7. We gave a proof by contrapositive:

Proof 1 If $a > 1$ and $b > 1$, then $ab > a > 1$, and $ab \neq 1$.

We may give a proof by contradiction.

Proof 2 Suppose there are real numbers a and b such that $ab = 1$, and $a > 1$ and $b > 1$. Then $ab > a > 1$, contradicting that $ab = 1$.

Both proofs have almost the same steps. We should use contrapositive; it is simpler and more direct than a proof by contradiction, which starts with setting up something false and then debunking it!